# Graphs

# Use of Graphs

- Shortest route between two cities.
- Shortest distance between all pairs of cities.
- Matching / Resource Allocation
- Task scheduling

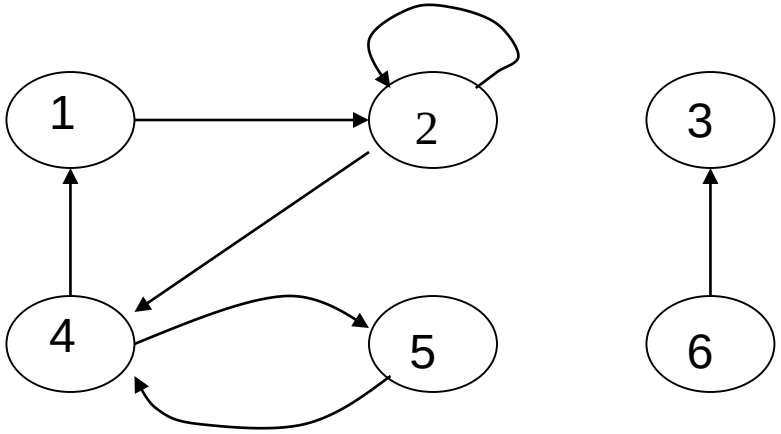
# Famous problems

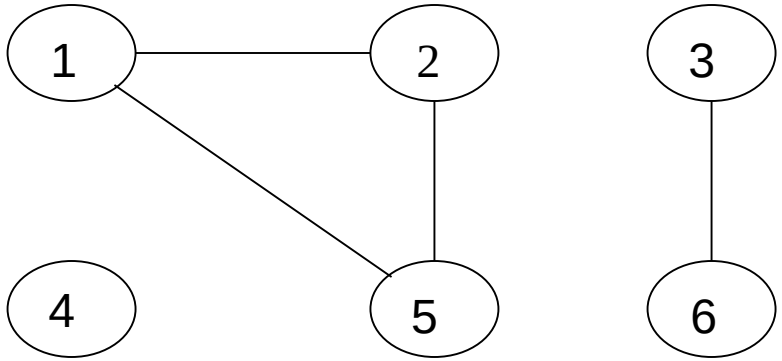
- The traveling salesman problem”

  A traveling salesman is to visit a number of cities; how to plan the trip so every city is visited once and just once and the whole trip is as short as possible ?

# Graphs

- Informally a *graph* is a set of nodes joined by a set of lines or arrows.

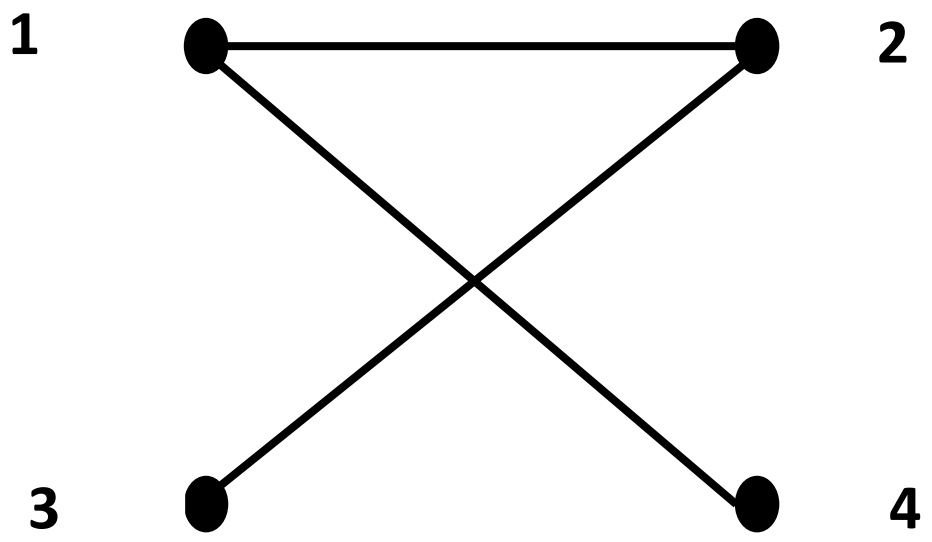


# Graphs

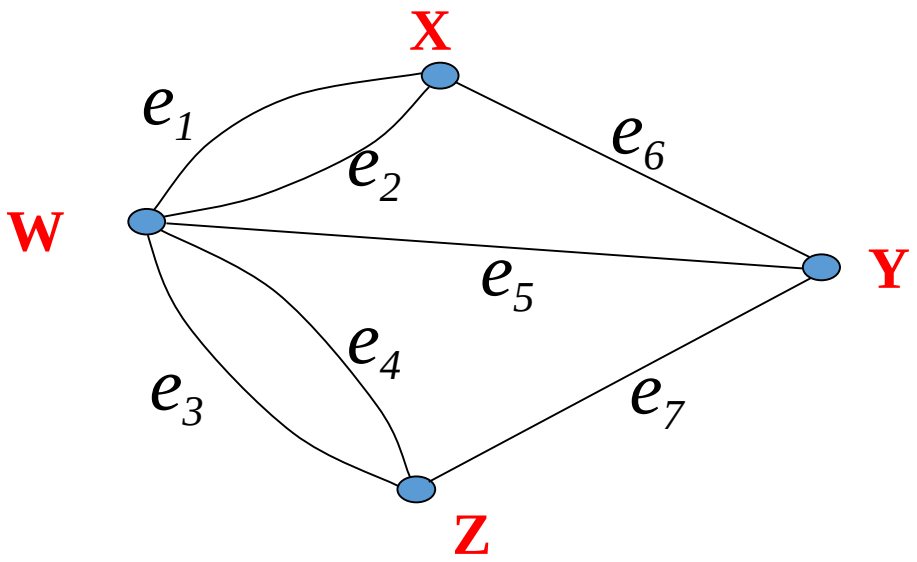
- Graphs is special data structure, which we can represent through two sets-
  - Set of Vertices / Nodes - V
  - Set of Edges - E

- Let,
  - V = {1, 2, 3, 4}
  - E = {(1,2), (2,3), (1,4)}

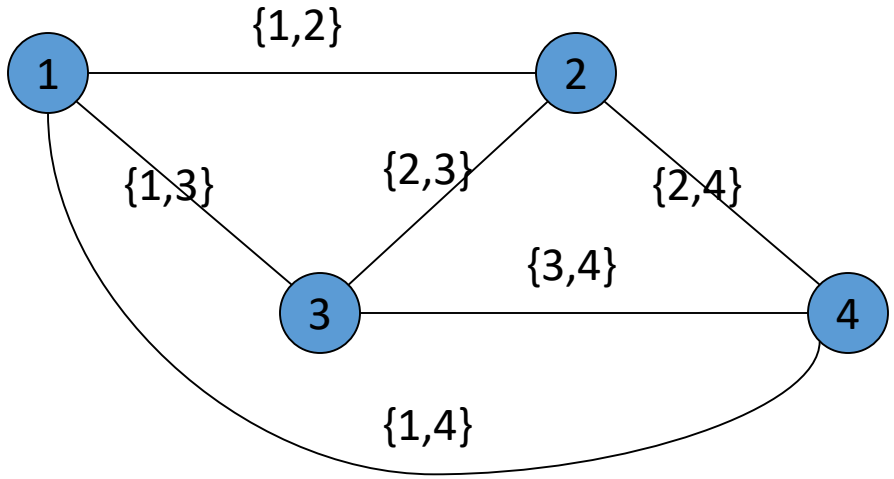


# Graphs

- A graph $G$ is a set consisting of:
  - A vertex set $V(G)$
  - An edge set $E(G)$



# Graphs Example




SET OF VERTICES

V = { 1, 2, 3, 4 }

SET OF EDGES

E = { (1, 2), (1, 3), (2, 3), (2, 4), (3, 4), (1, 4) }

# Infinite and Finite Graphs

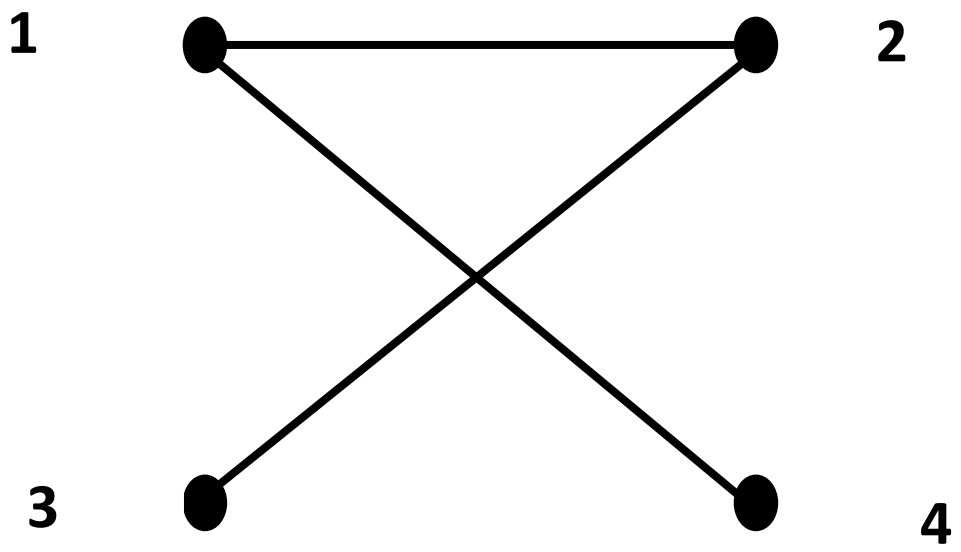
## Infinite Graphs

A graph with an infinite vertex set or an infinite number of edges is called an infinite graph.

## Finite Graphs

A graph with a finite vertex set and a finite edge set is called a finite graph.

# Simple Graphs Example



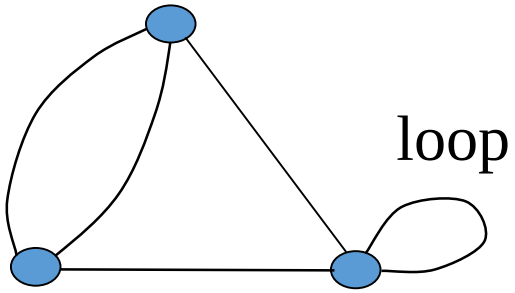

SET OF VERTICES

SET OF EDGES

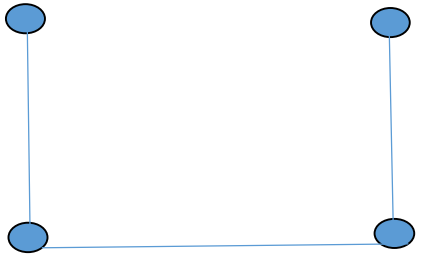
# Simple Graphs Example

***Simple graph*** **: A graph has no loops or multiple edges**

Multiple edges




It is not simple.

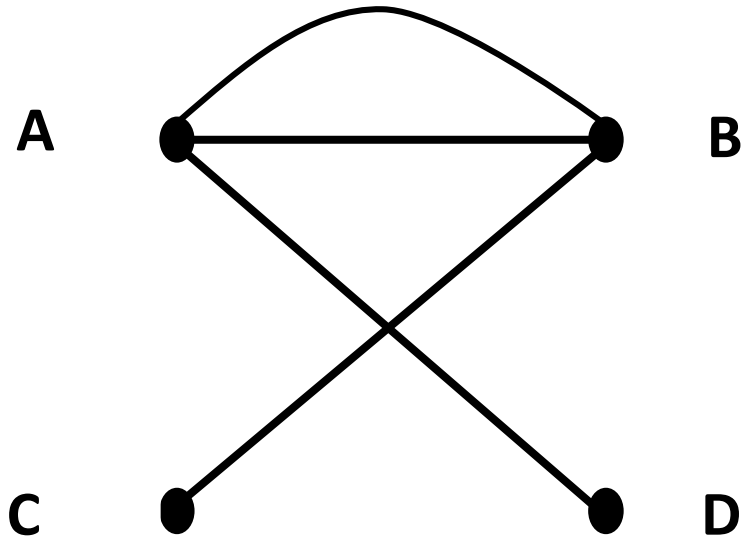


It is a simple graph.

# Multigraphs

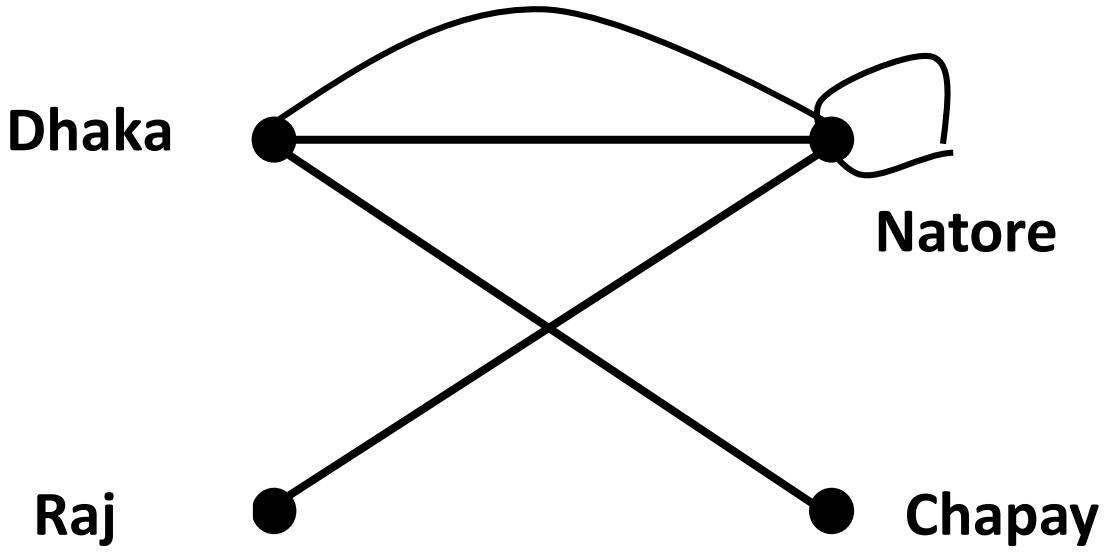
## Definition

Graphs that may have multiple edges connecting the same vertices are called multigraphs.



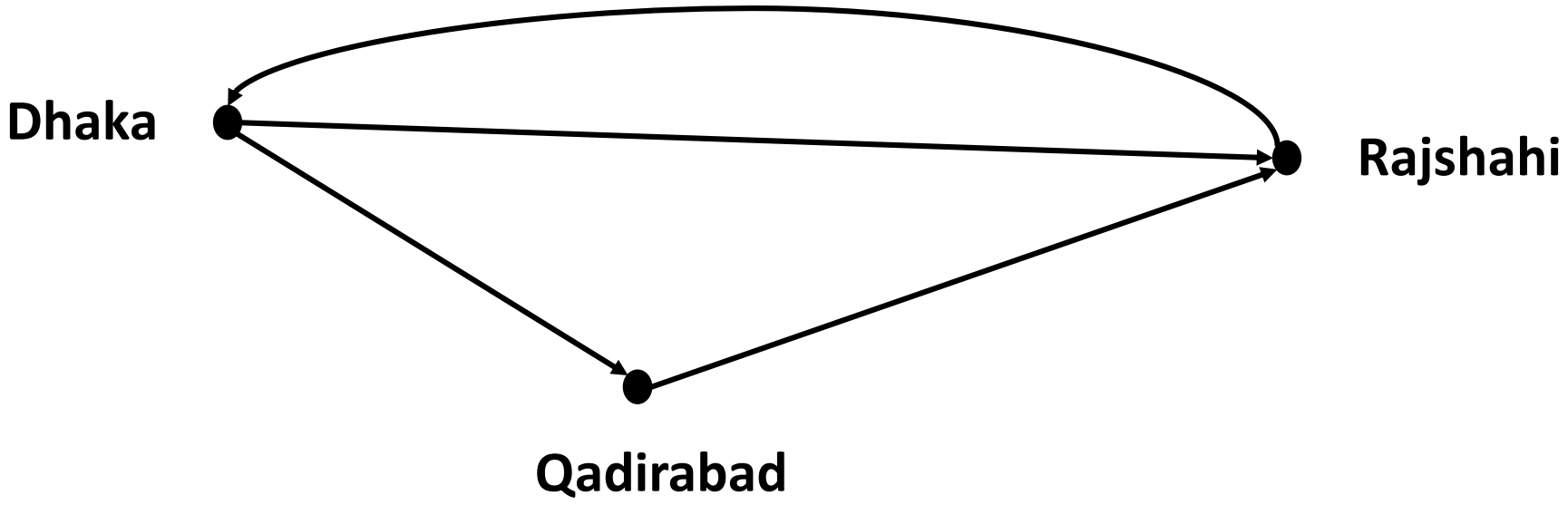
# Undirected Graphs

## Definition



# Directed Graphs

- Dhaka to Rajshahi
- Dhaka to Qadirabad
- Rajshahi to Dhaka
- No Qadirabad to Dhaka, but Qadirabad to Rajshahi



# A Directed Graph

## Definition

A directed graph (or digraph) *(V, E)* consists of a nonempty set of vertices *V* and a set of directed edges (or arcs) *E*. Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start* at u and *end* at v.

The edge $(a,b)$ is also denoted by $a \rightarrow b$

- $a$ is called the ***source*** of the edge while
- $b$ is called the ***target*** of the edge.

# Directed Graphs

- Simple Directed Graphs
- Multiple Directed Graphs
- Mixed Graphs

# Undirected Graphs: Adjacent

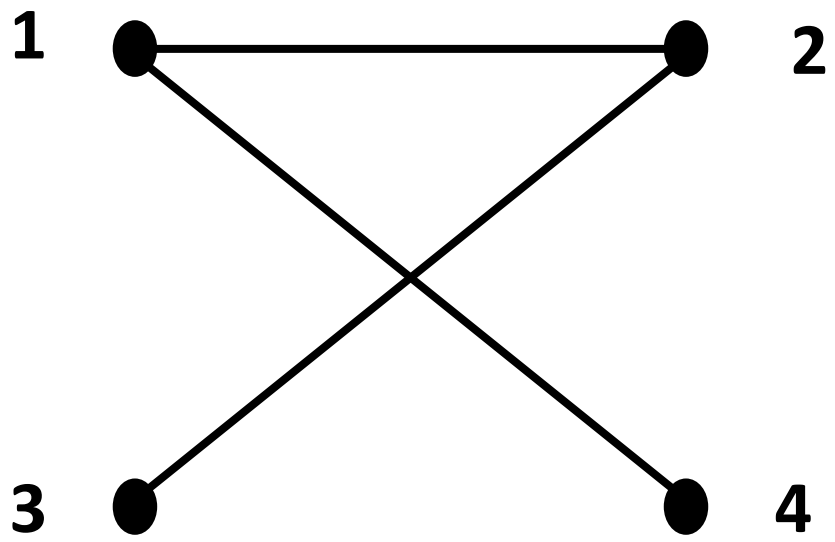
**Definition**

Vertices are ***adjacent*** if they are the endpoints of the same edge or they are connected by the same edge.

Adjacent of 1 : 2 and 4

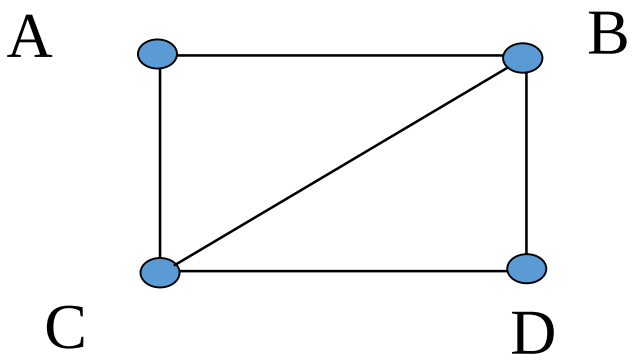
Adjacent of 2 : 1 and 3

Adjacent of 3 : 2



# Undirected Graphs: Adjacent

- Two vertices are *adjacent* and are *neighbors* if they are the endpoints of an edge
- Example:
  - *A* and *B* are adjacent
  - *A* and *D* are not adjacent

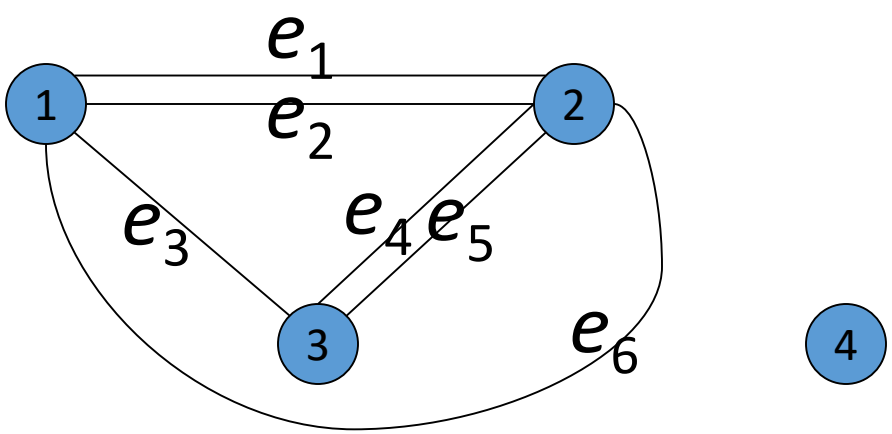


# Undirected Graphs Terminology Adjacent Vertices (Neighbors)

**Definition 1.** Two vertices, $u$ and $v$ in an undirected graph G are called **adjacent** (or **neighbors**) in G, if $\{u, v\}$ is an edge of G.

An edge $e$ connecting $u$ and $v$ is called **incident with vertices $u$ and $v$**, or is said to connect $u$ and $v$. The vertices $u$ and $v$ are called **endpoints** of edge $\{u, v\}$.

# Undirected Graphs Terminology




A: 1 is adjacent to 2 and 3

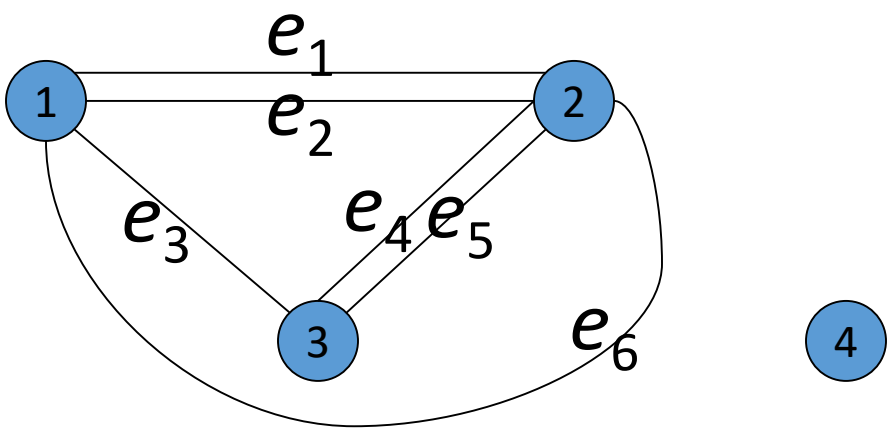
2 is adjacent to 1 and 3

3 is adjacent to 1 and 2

4 is not adjacent to any vertex

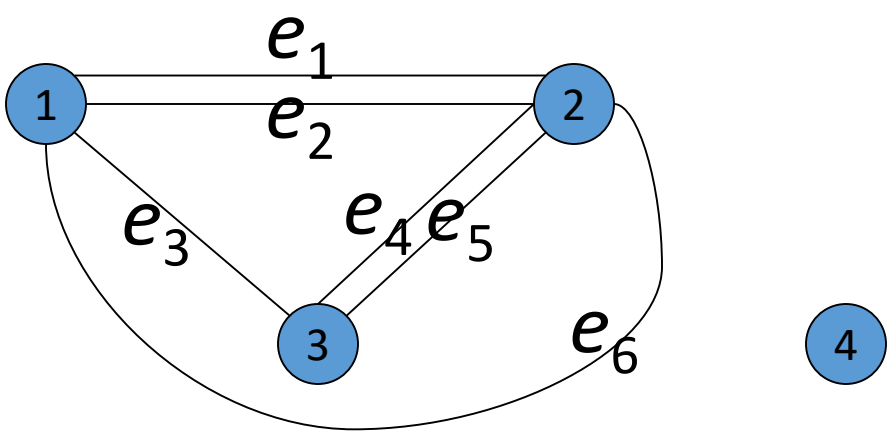
# Undirected Graphs Terminology

A vertex is ***incident*** with an edge (and the edge is incident with the vertex) if it is the endpoint of the edge.




Q: Which edges are incident to 1? How about incident to 2, 3, and 4?

# Undirected Graphs Terminology




A: $e_1$, $e_2$, $e_3$, $e_6$ are incident with 1

2 is incident with $e_1$, $e_2$, $e_4$, $e_5$, $e_6$

3 is incident with $e_3$, $e_4$, $e_5$

4 is not incident with any edge

# Undirected Graphs: Degree

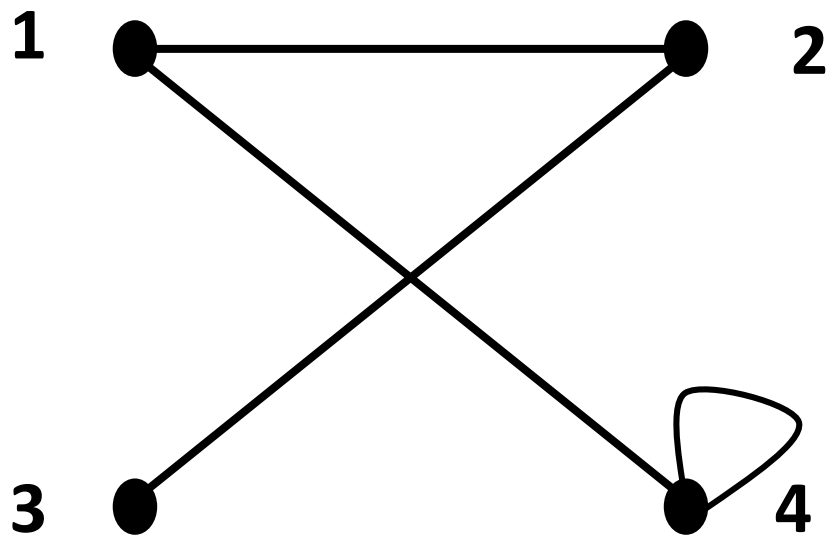
## Definition

The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by deg(v).

deg (1) : 2

deg (3) : 1

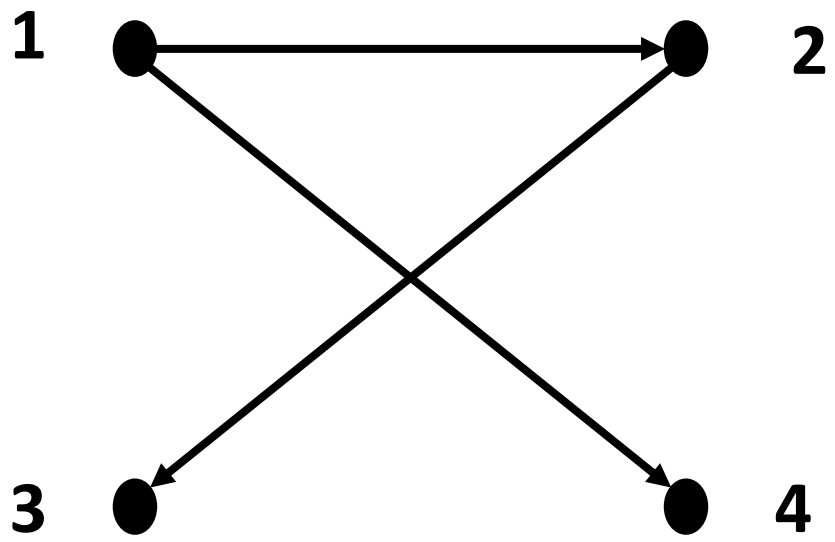
deg (4) : 3



# Directed Graphs : Degree

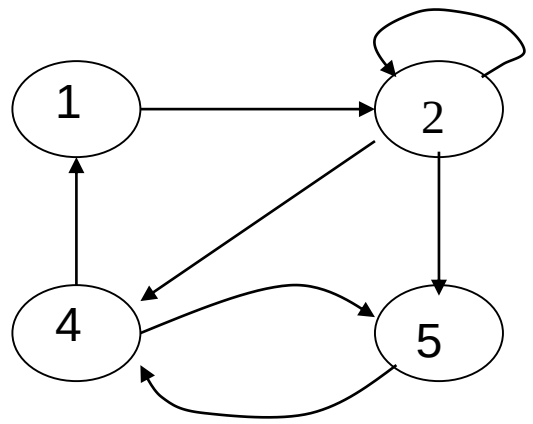
In degree :

Out degree:



# Directed Graphs : Degree

- In degree: Number of edges entering
- Out degree: Number of edges leaving




The in degree of 2 is 2 and the out degree of 2 is 3.

- Degree = indegree + outdegree

# Directed Graphs : Degree

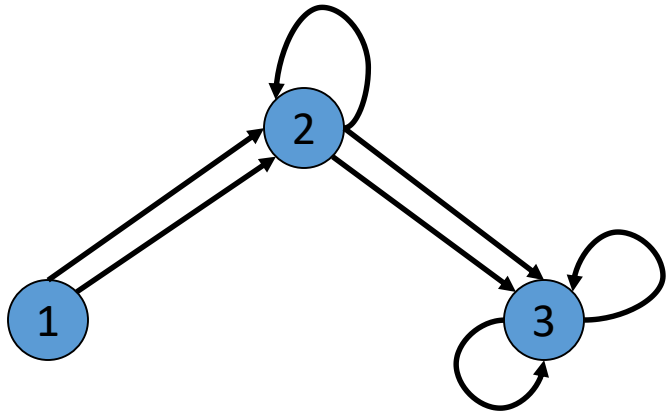
## Definition

In a graph with directed edges the *in-degree of a vertex v*, denoted by $\deg^-(v)$, is the number of edges with $v$ as their terminal vertex. The *out-degree of v*, denoted by $\deg^+(v)$, is the number of edges with $v$ as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

# Degree when Edges Directed

The ***in-degree*** of a vertex ($\deg^-$) counts the number of edges that stick *in* to the vertex.

The ***out-degree*** ($\deg^+$) counts the number sticking *out*.




Q: What are in-degrees and out-degrees of all the vertices?

# Oriented Degree when Edges Directed

A: $\deg^-(1) = 0$

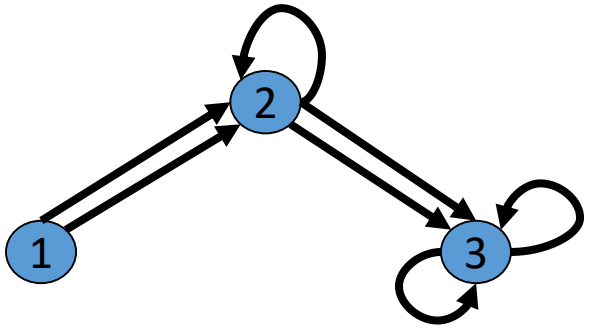
$\deg^-(2) = 3$

$\deg^-(3) = 4$

$\deg^+(1) = 2$

$\deg^+(2) = 3$

$\deg^+(3) = 2$



# Feedback

At last Class:

1. Definition Of Graphs
   - Simple Graphs
   - Infinite and Finite Graphs
   - Multigraphs
   - Directed Graphs
2. Adjacent
3. Degree
   - In degree
   - Out degree

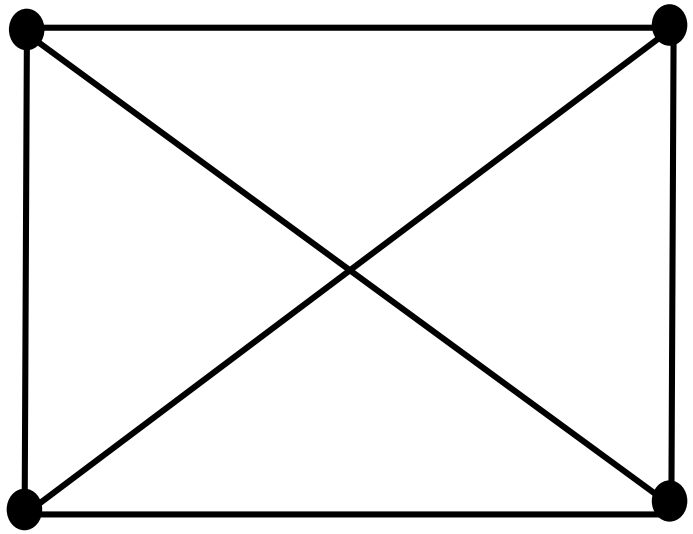
In this Class:

1. Special Simple Graphs
   - Complete Graphs
   - Cycles
   - Wheels
   - Regular graph
2. Bipartite
3. Handshaking Theorem
4. Adjacency Matrix

# Some Special Simple Graphs

## Complete Graphs - $K_n$

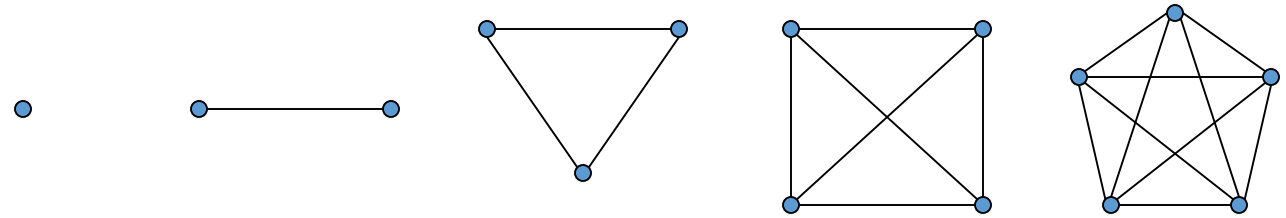
A complete graph on n vertices, denoted by $K_n$, is a simple graph that contains exactly one edge between each pair of distinct vertices.



No. of Edges = $n(n-1) / 2$

# Graph Patterns
# Complete Graphs - Kn

Complete Graphs – $K_n$ , when n = 1, 2, 3, 4, 5

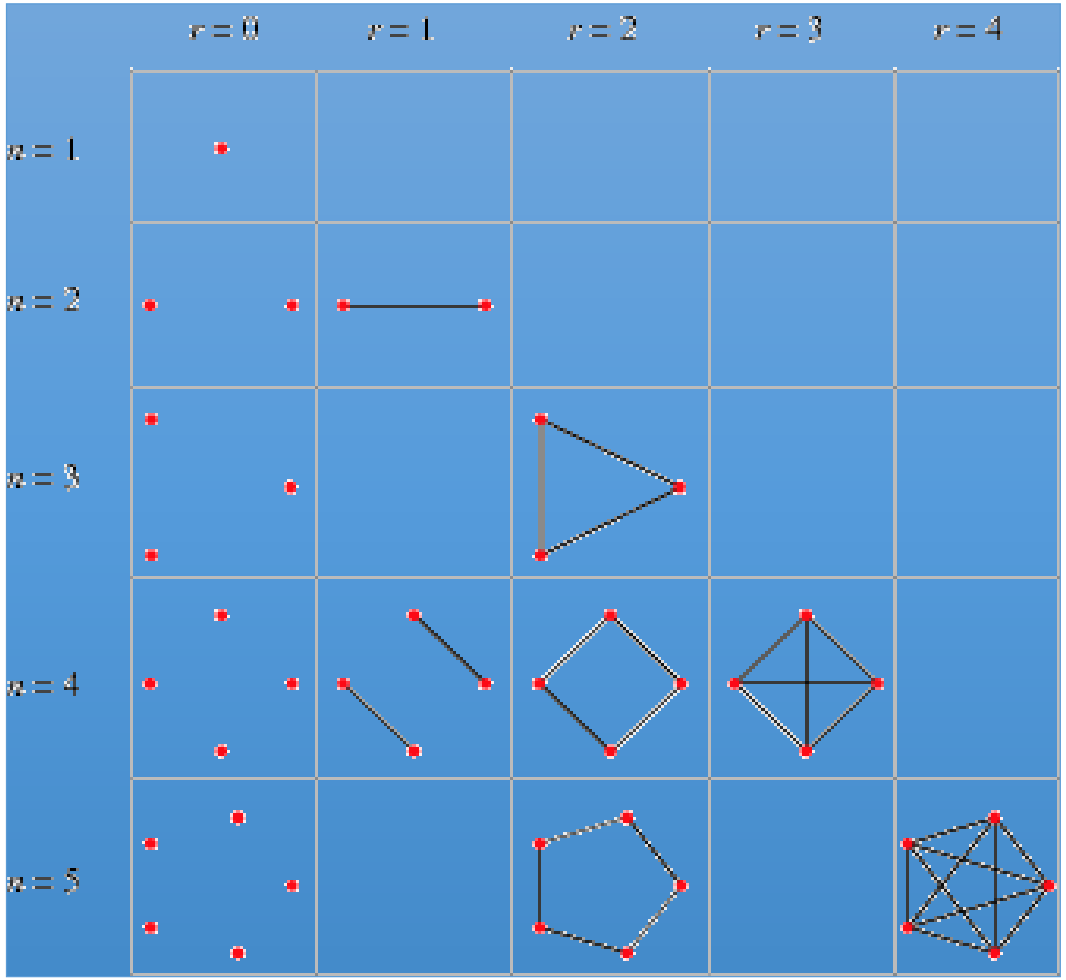


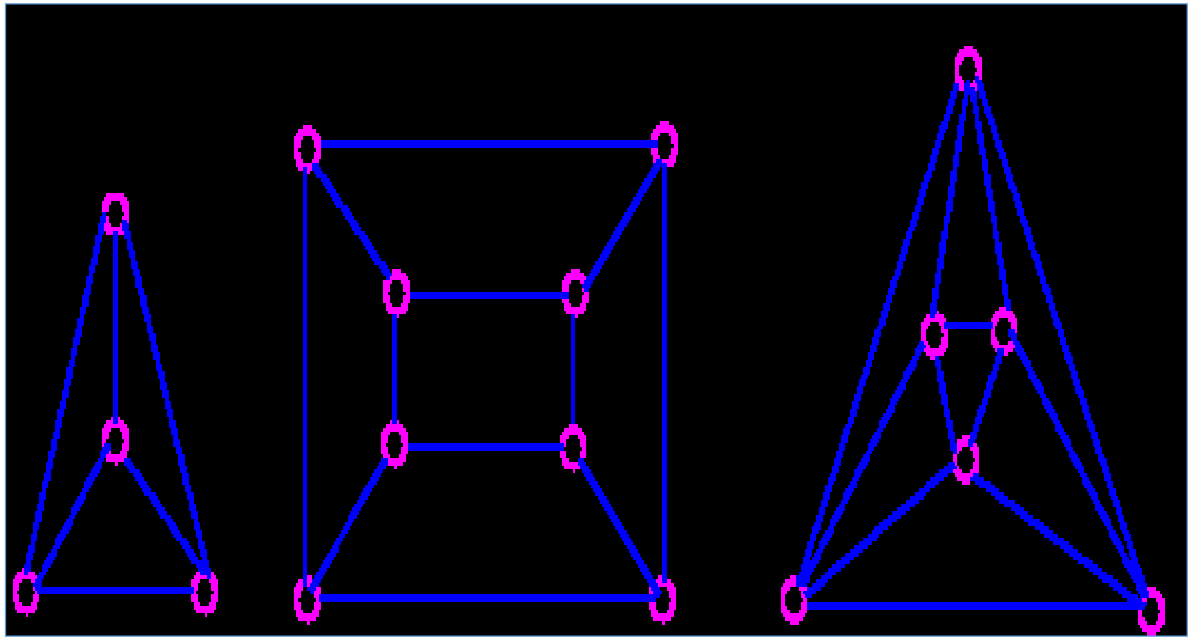
$K_1$ $K_2$ $K_3$ $K_4$ $K_5$

# Regular graph

A graph $G$ is *regular of degree $k$* or *$k$-regular* if every vertex has degree $k$. In other words, a graph is regular if every vertex has the same degree.

# Graph Patterns Regular graph



# Some Special Simple Graphs

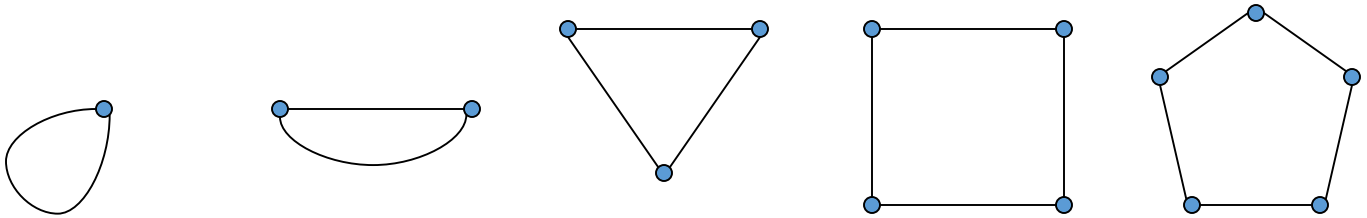
## Cycles - $C_n$

A cycle $C_n$, $n \geq 3$, consists of n vertices $v_1, v_2, \ldots\ldots, v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}, \ldots, \{v_n-1, v_n\}$, and $\{v_n, v_1\}$.

No. of Edges = n

# Graph Patterns
# Cycles - Cn

Q: What type of graph are $C_1$ and $C_2$ ?

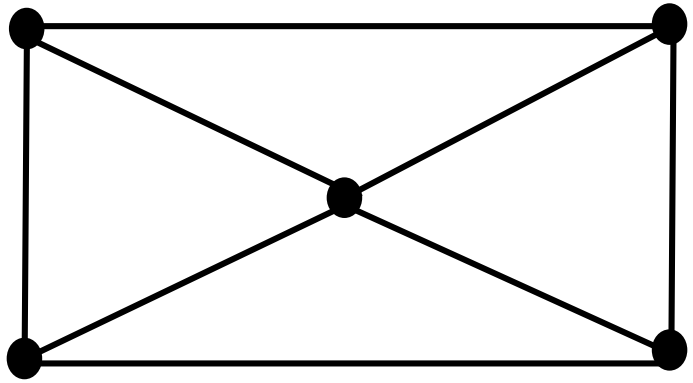


$C_1$ $C_2$ $C_3$ $C_4$ $C_5$

# Some Special Simple Graphs

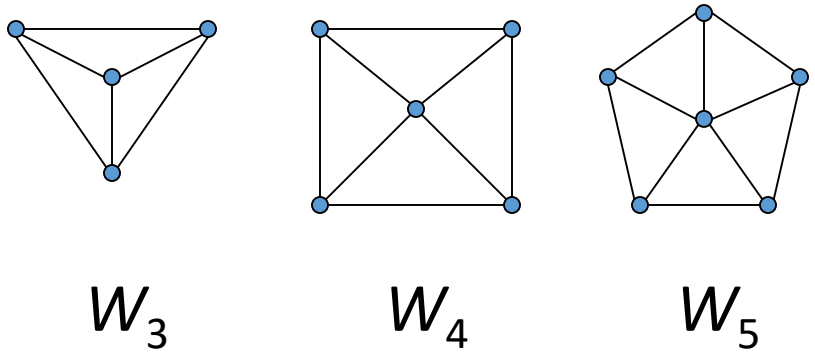
## Wheel Graph

We obtain a wheel $W_n$ when we add an additional vertex to a cycle $C_n$, for $n \geq 3$, and connect this new vertex to each of then vertices in $C_n$, by new edges.



# Graph Patterns
# Wheels - Wn

$W_n$ ; n = 3, 4, 5

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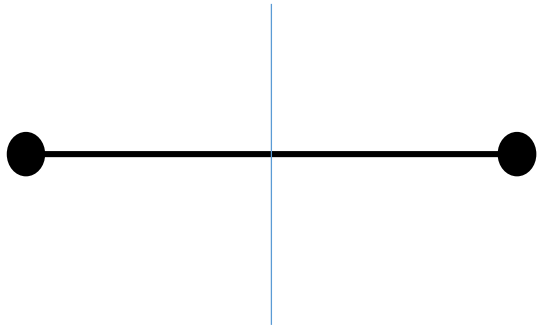
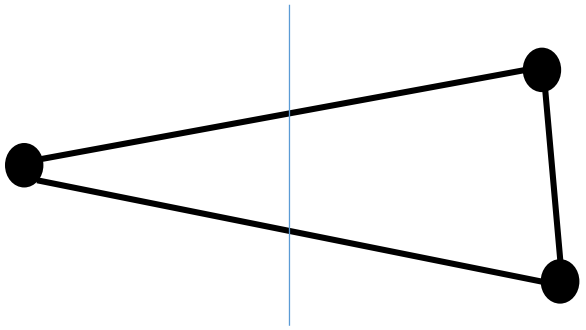
# Bipartite Graphs

## Definition

A simple graph $G$ is called *bipartite* if its vertex set $V$ can be partitioned into two disjoint sets $V_1$ and $V_2$ such that every edge in the graph connects a vertex in $V_1$ and a vertex in $V_2$ (so that no edge in $G$ connects either two vertices in $V_1$ or two vertices in $V_2$). When this condition holds, we call the pair $(V_1, V_2)$ a *bipartition* of the vertex set $V$ of $G$.

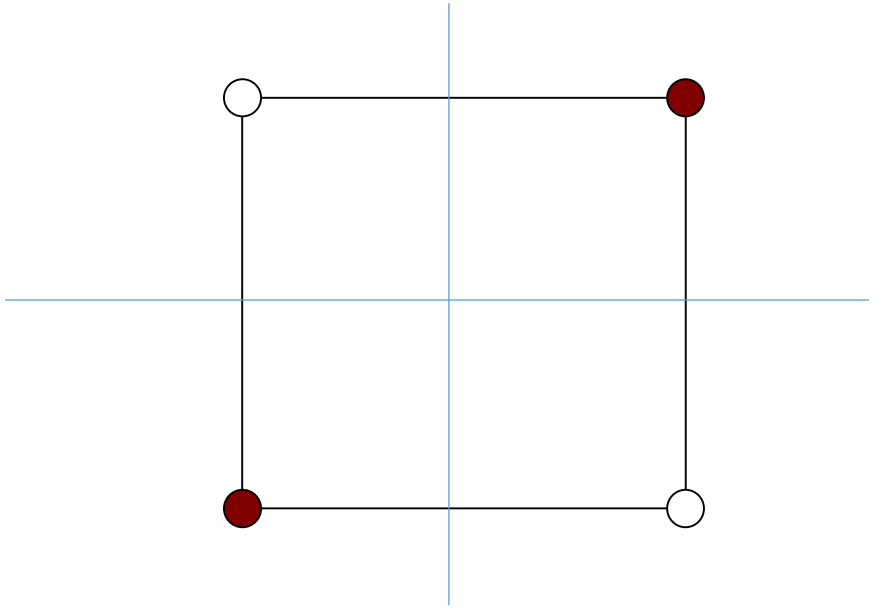




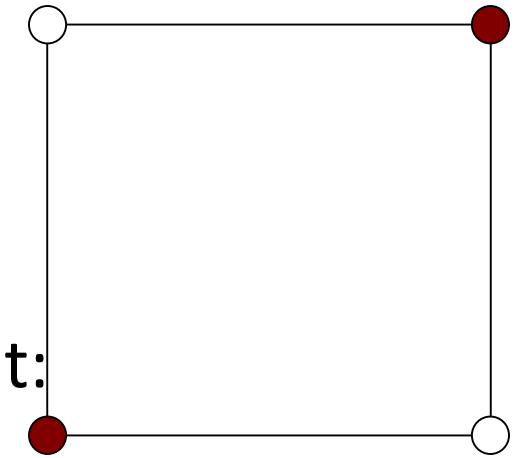
# Bipartite Graphs

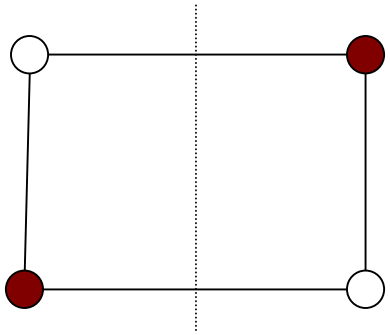
## Another definition

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

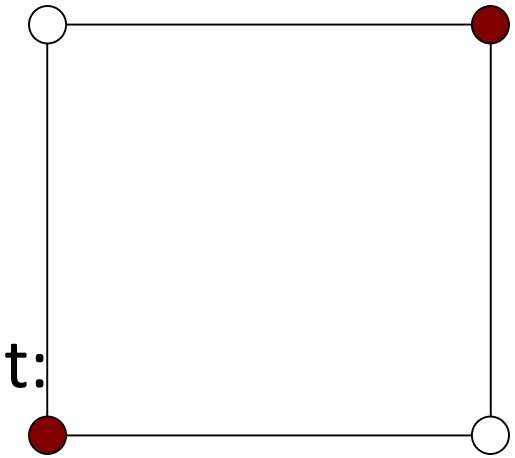


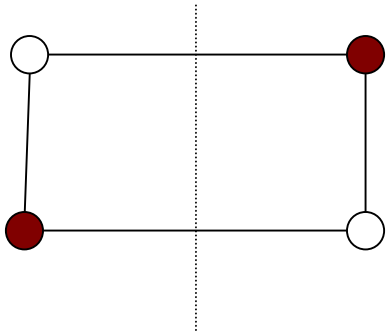
And so is bipartite, if we redraw it:



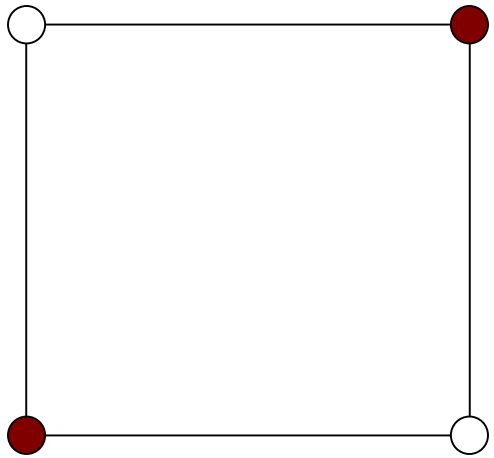


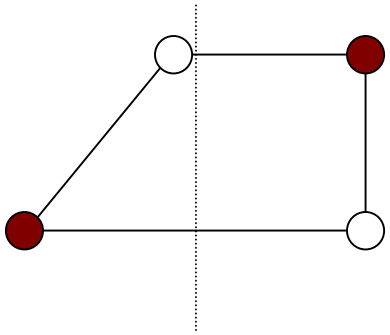
And so is bipartite, if we redraw it:



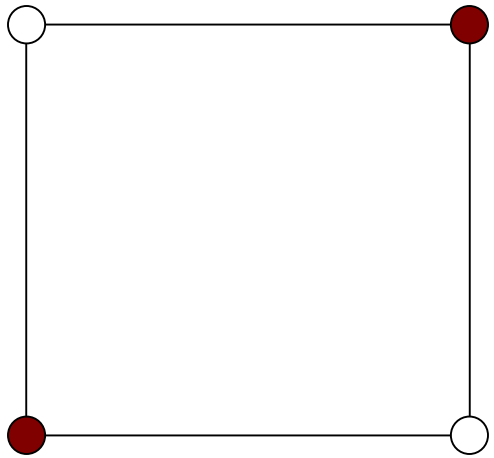


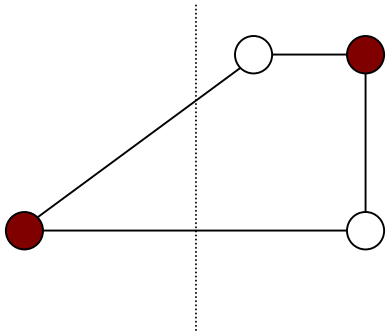
we redraw it:



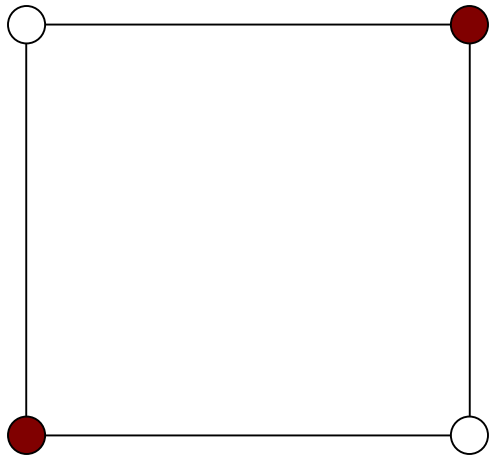


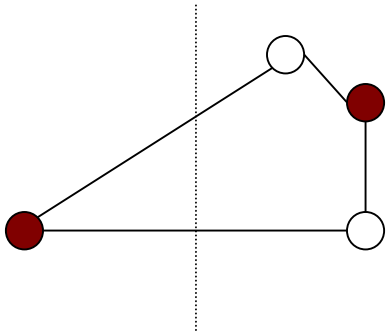
we redraw it:



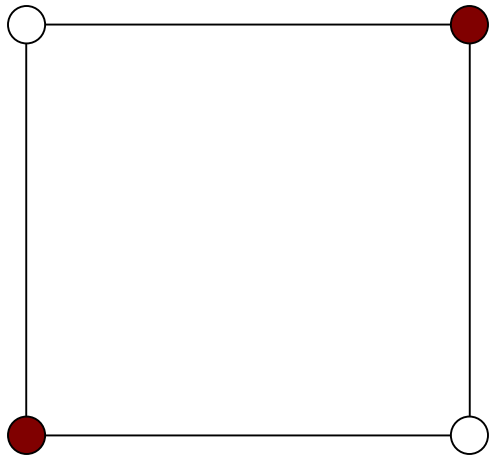


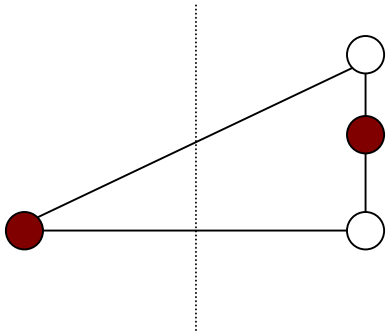
we redraw it:



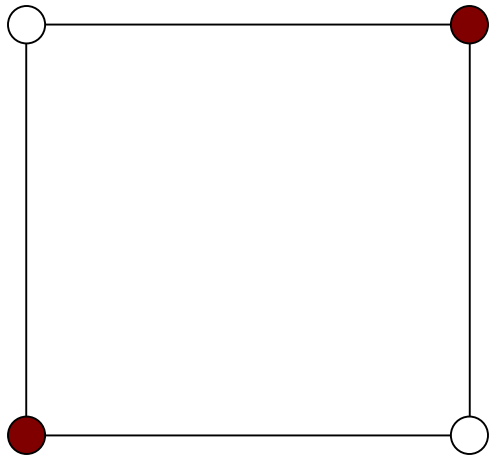


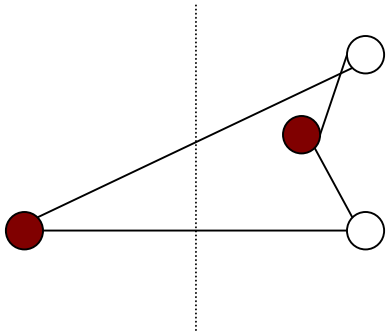
we redraw it:



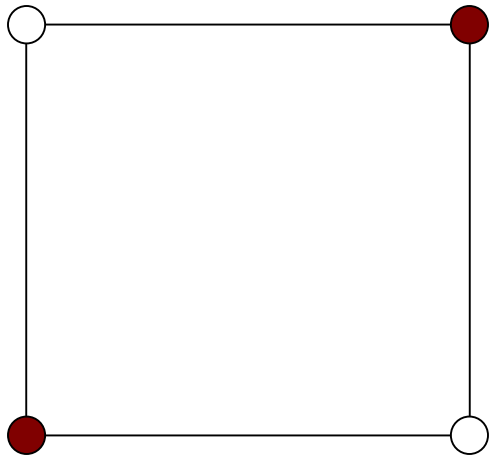


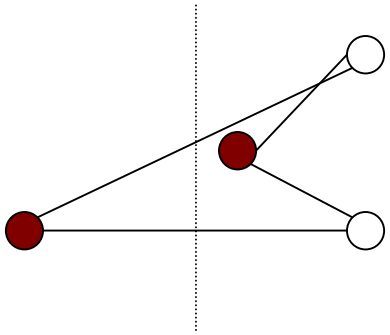
we redraw it:



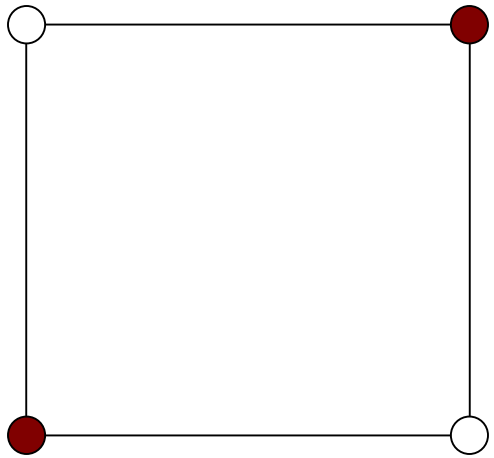


we redraw it:

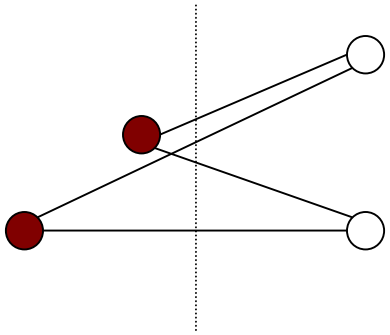




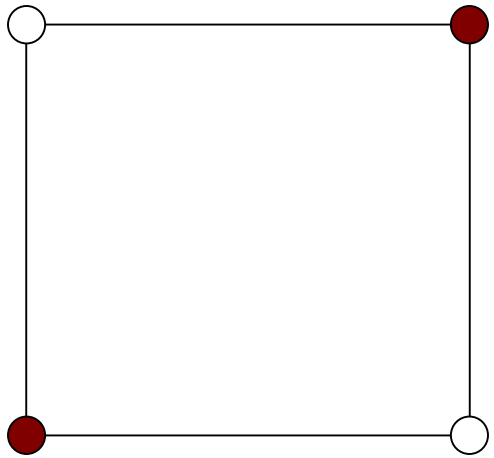
EG: $C_4$ is a bichromatic:

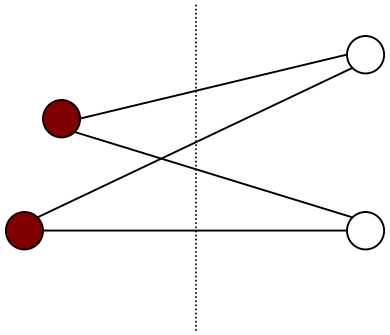


And so is bipartite, if we redraw it:

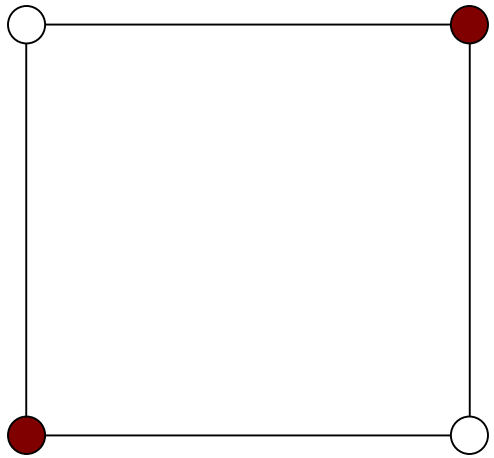


we redraw it:

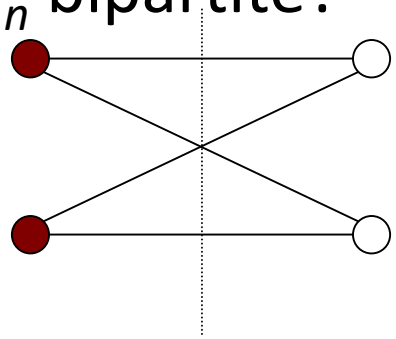




we redraw it:



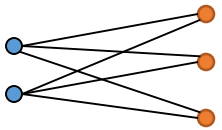
Q: For which $n$ is $C_n$ bipartite?



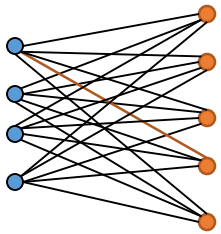
# Graph Patterns
# Complete Bipartite - Km,n

When all possible edges exist in a simple bipartite graph with *m* red vertices and *n* green vertices, the graph is called ***complete bipartite*** and the notation $K_{m,n}$ is used. EG:

$K_{2,3}$

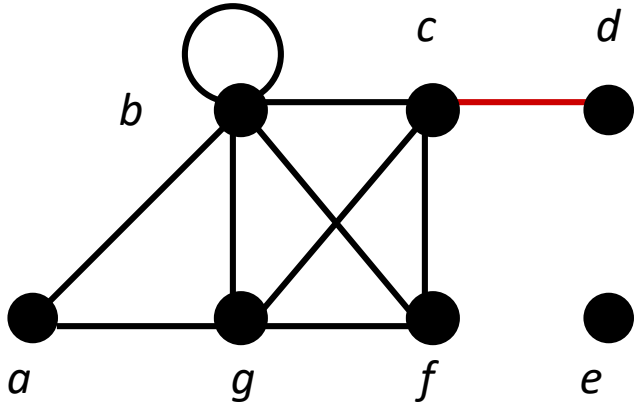


$K_{4,5}$



# Degree of a vertex

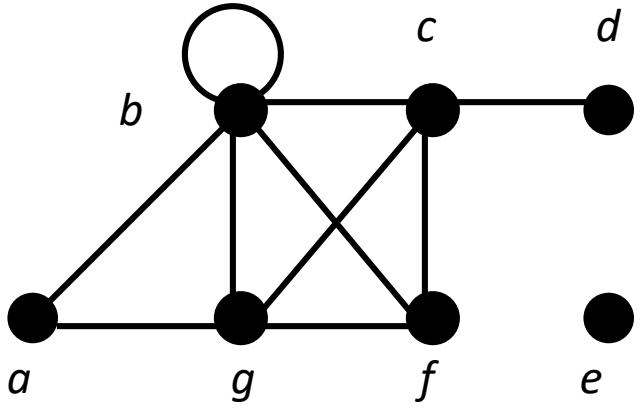
**Definition 1.** The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.




deg( *d* ) = 1

# Degree of a vertex

**Definition 1.** The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

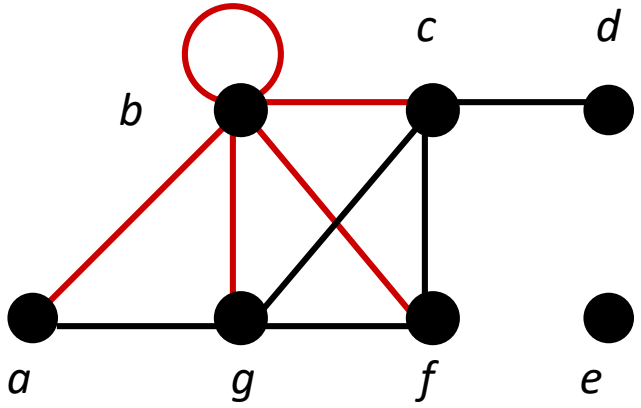



deg( *e* ) = 0

# Degree of a vertex

**Definition 1.** The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

deg( $b$ ) = 6

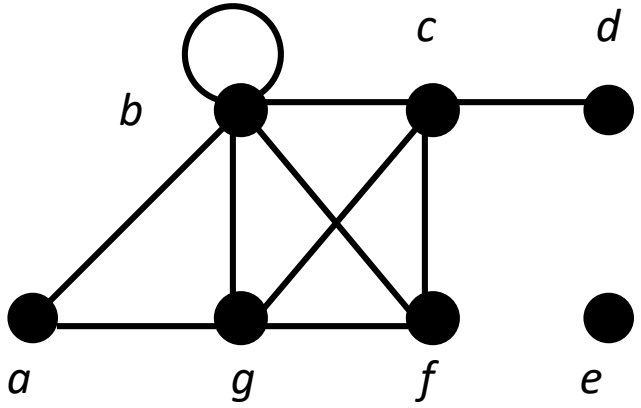


# Degree of a vertex

**Find the degree of all the other vertices.**

**deg( *a* )** **deg( c )** **deg( f )** **deg( *g* )**

deg( *b* ) = 6




deg( *d* ) = 1

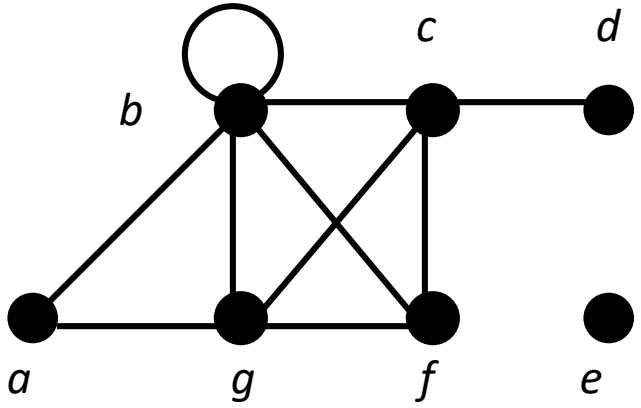
deg( *e* ) = 0

# Degree of a vertex

**Find the degree of all the other vertices.**

**deg( *a* ) = 2** **deg( c ) = 4** **deg( f ) = 3** **deg( *g* ) = 4**

deg( *b* ) = 6




deg( *d* ) = 1

deg( *e* ) = 0

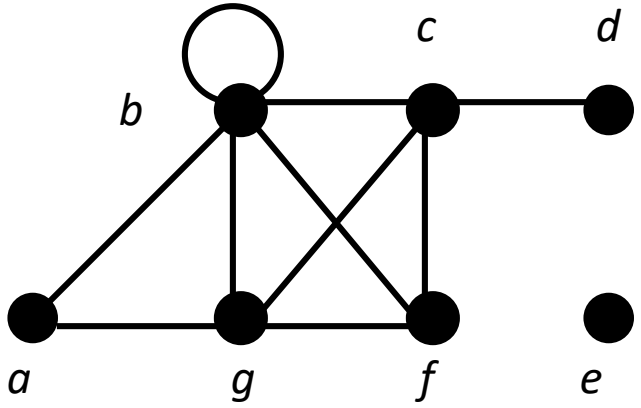
# Degree of a vertex

**Find the degree of all the other vertices.**

**deg( *a* ) = 2** **deg( c ) = 4** **deg( f ) = 3** **deg( *g* ) = 4**

**TOTAL of degrees = 2 + 4 + 3 + 4 + 6 + 1 + 0 = 20**

deg( *b* ) = 6




deg( *d* ) = 1

deg( *e* ) = 0

# Degree of a vertex

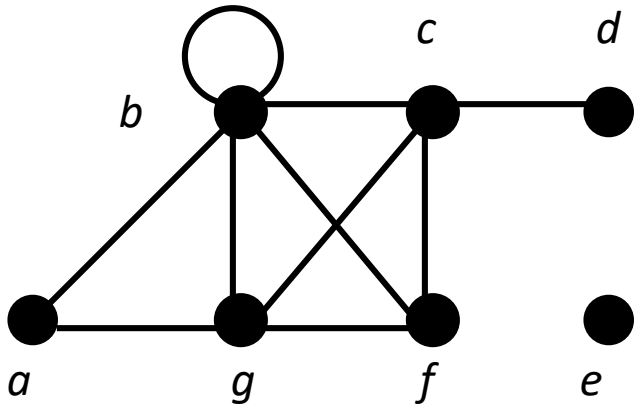
**Find the degree of all the other vertices.**

**deg( *a* ) = 2** **deg( c ) = 4** **deg( f ) = 3** **deg( *g* ) = 4**

**TOTAL of degrees = 2 + 4 + 3 + 4 + 6 + 1 + 0 = 20**

**TOTAL NUMBER OF EDGES = 10**

deg( *b* ) = 6




deg( *d* ) = 1

deg( *e* ) = 0

# Handshaking Theorem

**Theorem 1.** Let $G = (V, E)$ be an undirected graph $G$ with $e$ edges. Then

$$\sum_{v \in V} \deg(v) = 2e$$

*"The sum of the degrees over all the vertices equals twice the number of edges."*

**NOTE: This applies even if multiple edges and loops are present.**

# Representation of Graph

- Adjacency Matrix
- Adjacency lists

# Adjacency Matrix

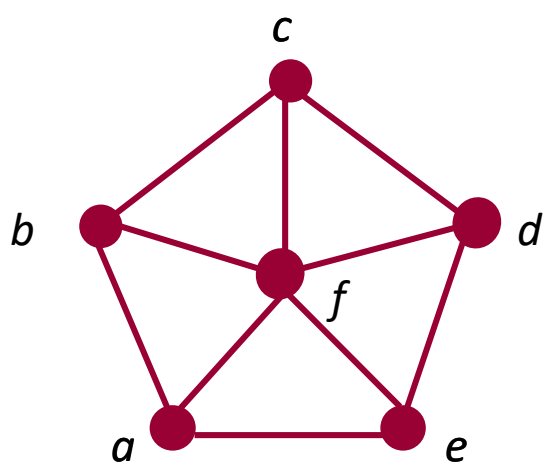
A simple graph $G = (V, E)$ with n vertices

can be represented by its adjacency matrix,

A, where entry $a_{ij}$ in row $i$ and column $j$ is

$$a_{ij} = \begin{cases} 1 & \text{if } \{ v_i, v_j \} \text{ is an edge in G,} \\ 0 & \text{otherwise.} \end{cases}$$

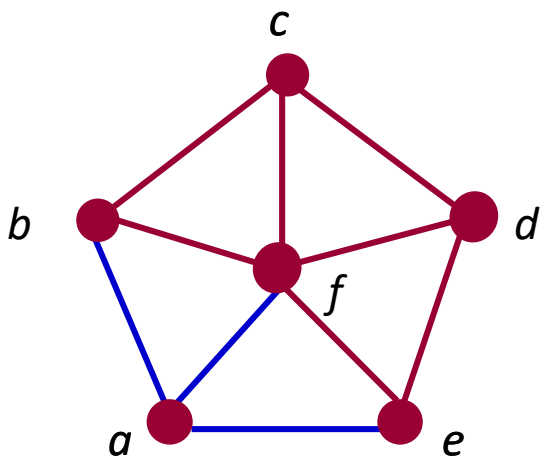
# Finding the adjacency matrix




$W_5$

**This graph has 6 vertices a, b, c, d, e, f. We can arrange them in that order.**

# Finding the adjacency matrix




$W_5$

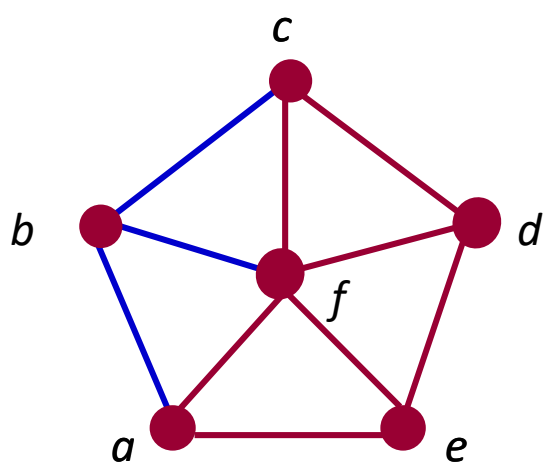
FROM

TO

$$\begin{array}{c|cccccc} & \mathbf{a} & \mathbf{b} & \mathbf{c} & \mathbf{d} & \mathbf{e} & \mathbf{f} \\ \hline a & 0 & 1 & 0 & 0 & 1 & 1 \\ b & & & & & & \\ c & & & & & & \\ d & & & & & & \\ e & & & & & & \\ f & & & & & & \end{array}$$

There are edges from a to b, from a to e, and from a to f

# Finding the adjacency matrix




$W_5$

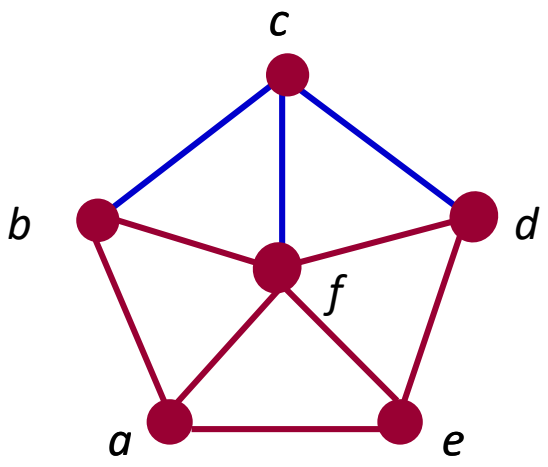
TO

FROM

$$
\begin{array}{c|cccccc}
 & \mathbf{a} & \mathbf{b} & \mathbf{c} & \mathbf{d} & \mathbf{e} & \mathbf{f} \\
\hline
a & 0 & 1 & 0 & 0 & 1 & 1 \\
b & 1 & 0 & 1 & 0 & 0 & 1 \\
c & & & & & & \\
d & & & & & & \\
e & & & & & & \\
f & & & & & &
\end{array}
$$

There are edges from b to a, from b to c, and from b to f

# Finding the adjacency matrix




$W_5$

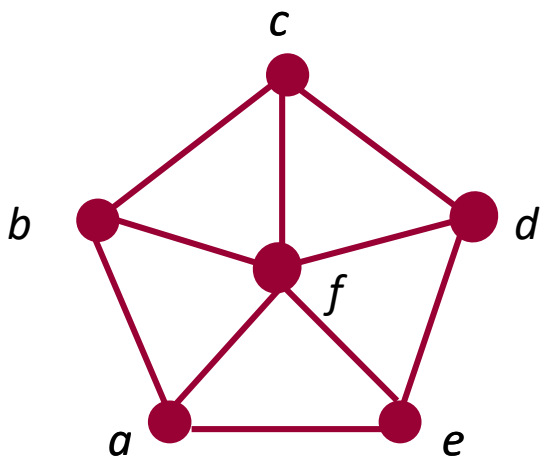
TO

FROM

$$\begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} \mathbf{a} & \mathbf{b} & \mathbf{c} & \mathbf{d} & \mathbf{e} & \mathbf{f} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

There are edges from c to b, from c to d, and from c to f

# Finding the adjacency matrix




$W_5$

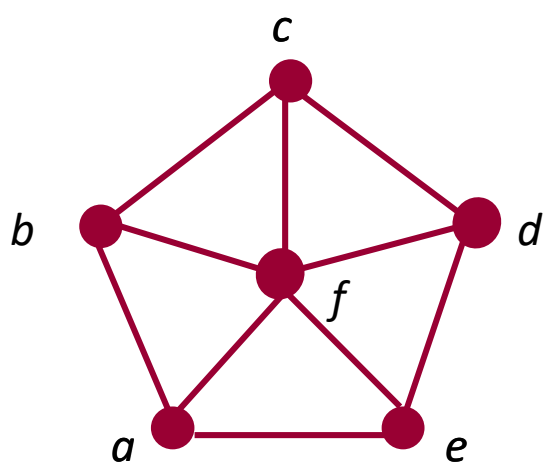
TO

FROM

$$\begin{array}{c|cccccc} & a & b & c & d & e & f \\ \hline a & 0 & 1 & 0 & 0 & 1 & 1 \\ b & 1 & 0 & 1 & 0 & 0 & 1 \\ c & 0 & 1 & 0 & 1 & 0 & 1 \\ d & & & & & & \\ e & & & & & & \\ f & & & & & & \end{array}$$

COMPLETE THE ADJACENCY MATRIX . . .

# Finding the adjacency matrix




$W_5$

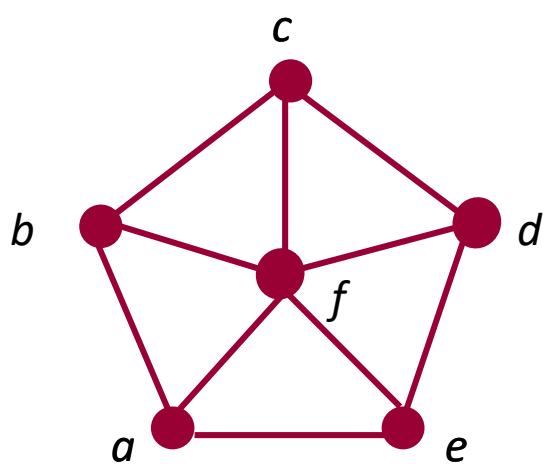
TO

FROM

$$\begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Notice that this matrix is symmetric. That is $a_{ij} = a_{ji}$ Why?

# Finding the adjacency List




$W_5$

TO

FROM

$$\begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

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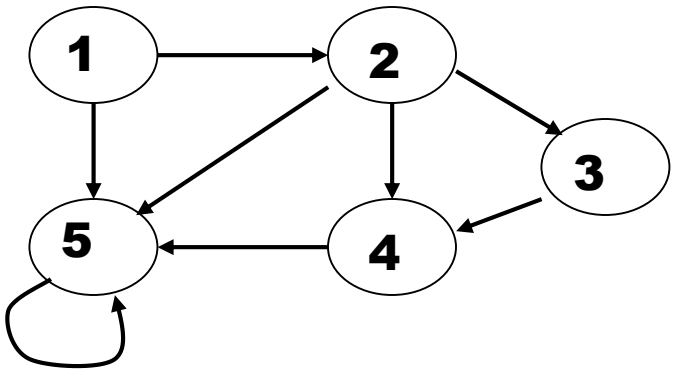
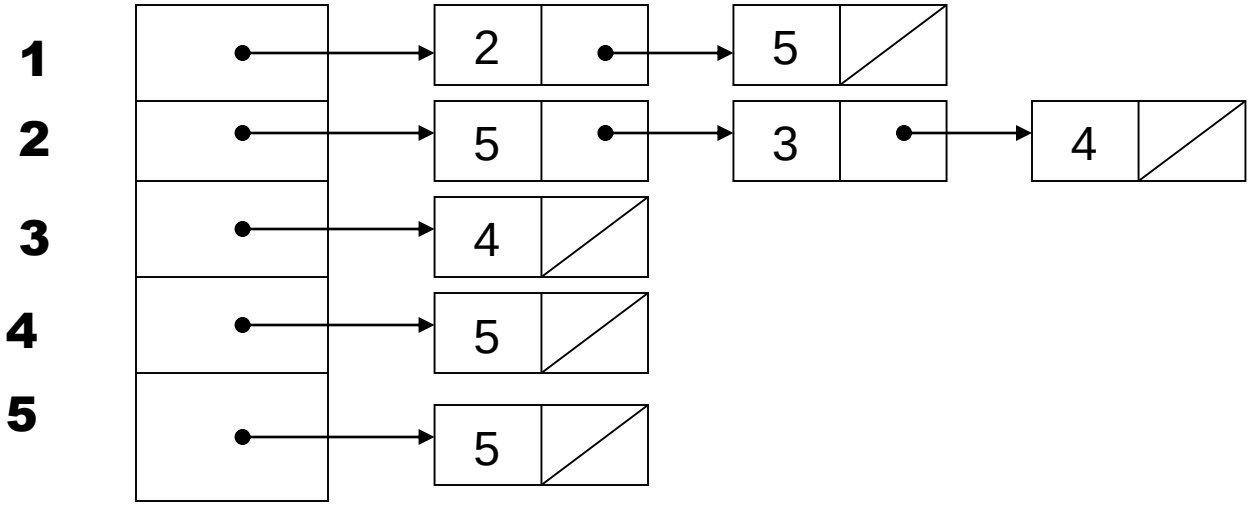
# Finding the adjacency List

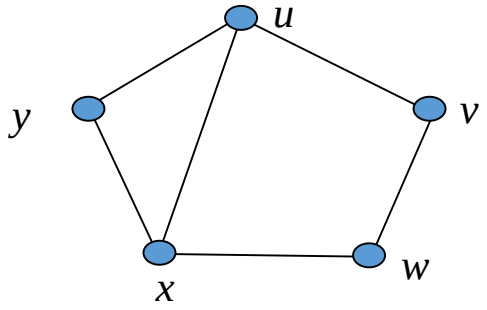


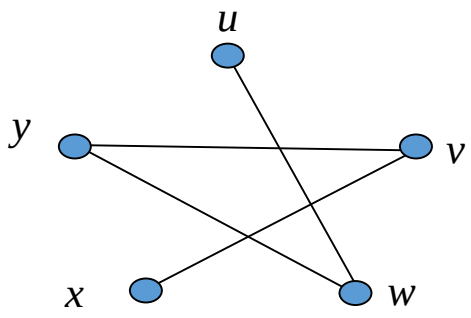
# Complement

- ***Complement of G***: The complement $G'$ of a simple graph $G$ :
  - A simple graph
  - $V(G') = V(G)$
  - $E(G') = \{ uv \mid uv \notin E(G) \}$

G




G'



# Feedback

In this Class:

1. Special Simple Graphs
   - Complete Graphs
   - Cycles
   - Wheels
   - Regular Graph
2. Bipartite
3. Handshaking Theorem
4. Adjacency Matrix
5. Adjacency List